Abdullah Mushtaq

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Short Bio

My primary research spans LLM inference, benchmarking, dynamic multi-agent systems, agentic planning, reducing LLM hallucination and sycophancy, alignment, and multimodal deep learning architectures. I also work on applied NLP, reinforcement learning, world models, and simulations. I have research experience and collaborations with internationally recognized top institutions, including Oxford, Weill Cornell Medicine, Imperial College London, Zayed University (Abu Dhabi), Qatar University, Copenhagen Business School, LUMS, and ITU Lahore. I have authored multiple peer-reviewed publications at top-tier venues, including the Journal of Artificial Intelligence Research (JAIR, Q1, H-index: 143) and NeurIPS 2025, with additional work under review at Artificial Intelligence and Law. I have led end-to-end projects from dataset and system design through to peer-reviewed publication. The media and national TV have covered my research. I have served as a TA for three courses at ITU, supporting up to 250+ undergraduate and graduate students. I am a recipient of the nationally competitive HEC Ehsaas fully-funded undergraduate scholarship and the NGIRI 2023–24 national ICT research grant (IGNITE), and I graduated in the top 5% of my class on the Dean's Honor List at ITU Lahore. Beyond research, I co-founded the ITU Games Society and served as Head of Game Arts and Design. I have also mentored 17 interns as Lab Lead at ITU, guiding junior researchers across multiple projects.

Education

Information Technology University — Lahore, Pakistan
Bachelor of Science in Computer Science — *2020 – 2024*
Dean's Honor List | Rank: Top 5% | Specialization GPA: 3.83/4.0 | Merit Scholarship 2020-2024

The Educators College — Chichawatni, Pakistan
Intermediate Education, Intermediate in Computer Science | Grade: A+ | Topper in Computer Science — *2018 – 2020*

Dawn High School — Chichawatni, Pakistan
Secondary Education | Grade: A+ | District Topper in Computer Science & Physics; School Top Scorer in Urdu — *2016 – 2018*

Publications

[1] **Mushtaq, Abdullah**, Taj, Imran, Naeem, Rafay, Ghaznavi, Ibrahim, & Qadir, Junaid. (2026). WorldView-Bench: A benchmark for evaluating global cultural perspectives in large language models. *Journal of Artificial Intelligence Research*, Vol. 85, Page 28. (Acceptance Rate: 12.6%, Q1, H-index: 143) DOI: 10.1613/jair.1.19001

[2] **Mushtaq, Abdullah**, Naeem, Rafay, Elmahjub, Ezieddin, Ghaznavi, Ibrahim, Al-Maliki, Shawqi Mohammed, Abdallah, Mohamed, Al-Fuqaha, Ala, & Qadir, Junaid. Can LLMs Write Faithfully? An Agent-Based Evaluation of LLM-generated Islamic Content. In *5th Muslims in ML Workshop co-located with NeurIPS 2025*. arXiv:2510.24438

[3] Elmahjub, Ezieddin, Qadir, Junaid, **Mushtaq, Abdullah**, Naeem, Rafay, Ghaznavi, Ibrahim, & Iqbal, Waqar. (2026). IslamicLegalBench: Evaluating LLMs knowledge and reasoning of Islamic law across 1,200 years of Islamic pluralist legal traditions. *Artificial Intelligence and Law* (Under Review). arXiv:2602.21226

[4] **Mushtaq, Abdullah**, Naeem, Rafay, Ghaznavi, Ibrahim, Taj, Imran, Hashmi, Imran, & Qadir, Junaid. (2025, April). Harnessing multi-agent LLMs for complex engineering problem-solving: A framework for senior design projects. In *2025 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1–10). IEEE. arXiv:2501.01205

[5] **Mushtaq, Abdullah**, Naeem, Muhammad Rafay, Taj, Muhammad Imran, Ghaznavi, Ibrahim, & Qadir, Junaid. (2025). Toward inclusive educational AI: Auditing and alignment of frontier LLMs for cultural biases through a multiplexity lens. In *2025 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1–10). IEEE. arXiv:2501.03259

[6] **Mushtaq, Abdullah**., Naeem, Muhammad Rafay., Ghaznavi, Ibrahim, Abd-alrazaq, Alaa, Tabassum, Aliya, & Qadir, Junaid (2025). Can Agents Judge Systematic Reviews Like Humans? Evaluating SLRs with LLM-based Multi-Agent System. (In-progress) arXiv:2509.17240

Experience & Research Work

Qatar University — Doha, Qatar

Visiting Research Assistant — LLM Reasoning & Agentic AI — *Aug. 2025 – Present*
Supervisors: Ezieddin Elmahjub (Harvard & Qatar University), Junaid Qadir (Qatar University), Ibrahim Ghaznavi (ITU)
Project: IslamicLegalBench: Evaluating LLM Reasoning Across 1,200 Years of Islamic Law
AI & Law (submitted) | Paper | Leaderboard

- Constructed a multi-dimensional benchmark comprising 14 distinct task types, spanning basic factual recall to advanced jurisprudential reasoning, from Islamic legal texts across 1,200+ years and multiple schools of thought.
- Architected an end-to-end large-scale evaluation pipeline with reproducible protocols, semantic scoring, and statistical validation; rigorously evaluated frontier LLMs (GPT-5, Gemini 2.5, Claude 4, DeepSeek R1, Grok 4, Llama 4) under LLM-as-a-Judge, expert human annotation, and structured prompting strategies (Zero-shot, Few-shot, Chain-of-Thought).
- Serving as **lead contributor** on the manuscript, advancing the state of domain-specific legal reasoning in AI.

Project: SLR-GPT: Can LLM-Agents Judge Systematic Reviews Like Humans? ***In-progress*** | Paper
Project Collaborators: Alaa A. Abd-alrazaq (Weill Cornell Medicine), Aliya Tabassum (Qatar University

- Designed and built a Multi-Agent System of 27 specialized LLM-based agents organized into deliberative societies, integrating few-shot prompting, real-time tool use, MCP, Vision Language Models, and an ArXiv Toolkit into a unified multimodal agentic pipeline for comprehensive SLR analysis from a single PDF input.
- Engineered an interactive Flask-based interface enabling seamless agent-user interaction, structured paper analysis, and interpretable insight extraction at scale.
- Achieved **84%** agreement with domain experts across medicine, AI, and AR/VR systematic reviews, demonstrating strong reliability for AI-assisted scholarly review assessment.

Hamad Bin Khalifa University (HBKU) — Doha, Qatar (remote)

Research Assistant — Agentic AI & Large Language Models *May 2025 – Aug. 2025*
Supervisors: Mohamed Abdallah (HBKU), Ala Al-Fuqaha (HBKU), Shawqi M. Al-Maliki (HBKU), Junaid Qadir (Qatar University)
Project: Can LLMs Write Faithfully? — Agent-Based Evaluation of LLM-generated Religious Content
NeurIPS 2025 (MusIML) | Paper | Code

- Developed a dual-agent evaluation framework (quantitative + qualitative) to assess LLM reliability in producing religious guidance, incorporating citation verification, multi-dimensional scoring, and structured interpretability schemas at scale.
- Benchmarked frontier and domain-specific LLMs on authentic religious essay prompts, identifying critical gaps in citation accuracy, religious consistency, and faith-sensitive content generation.

Zayed University — Abu Dhabi, UAE

Research Assistant — Large Language Models *July 2024 – July 2025*
Supervisors: Imran Hashmi (Oxford), Junaid Qadir (Qatar University), Imran Taj (Zayed University)
Project: WorldView-Bench — Evaluating Global Cultural Perspectives in LLMs ***JAIR 2026*** | Paper | Code

- Introduced a benchmark of 175 questions across 7 categories for measuring cultural inclusivity in frontier LLMs; improved inclusivity metrics from **13%** to **94%** while reducing negative sentiment to **2.4%** via MAS-based multiplex approaches.

Project: MASC — Multi-Agent SDP Co-Pilot for Complex Engineering ***IEEE EDUCON 2025*** | Paper | Code

- Built a collective intelligence framework for automated assessment of engineering projects, achieving **89%** alignment with expert faculty evaluations; integrated advanced NLP metrics (Lexical Cohesion, Clause Density) and dynamic agent-based QnA.

Project: Cultural Bias Auditing — Alignment of Frontier LLMs ***IEEE EDUCON 2025*** | arXiv

- Developed methodologies for detecting and mitigating cultural biases in frontier LLMs (GPT-4, Claude 3.5, Llama 3.1 & 3.2, Mistral 7B), improving inclusivity rates from **3.25%** to **98%** via bias-aware prompt calibration, controlled generations, and Multi-Agent Systems.

Lahore University of Management Sciences (LUMS) — Lahore, Pakistan

Research Assistant — Natural Language Processing & Deep Learning *Aug. 2024 – Nov. 2024*
Supervisors: Ashish Gehani (Stanford Research Institute), Fareed Zaffar (LUMS)
Project: Deep Learning-based Intrusion Detection Systems — Evaluation & Robustness Analysis

- Evaluated and stress-tested deep learning-based intrusion detection systems (KAIROS, FLASH, ProvDetector) using Word2Vec, GCNs, and XGBoost, identifying implementation bugs, dataset limitations, and performance bottlenecks.
- Formalized reproducibility reports and proposed real-world bypass scenarios, contributing to robustness and reliability assessment of open-source security systems.

CSaLT Lab, Lahore University of Management Sciences (LUMS) — Lahore, Pakistan

Undergraduate Research Assistant — Natural Language Processing *Aug. 2023 – Apr. 2024*
Supervisors: Agha Ali Raza (LUMS & CMU), Abbas Edalat (Imperial College London)
Project: Urdu-language Psychotherapy Chatbot — Low-resource NLP for Mental Health *Joint CSaLT–Imperial College London*

- Fine-tuned and integrated state-of-the-art models (Whisper, M4T, BERT, MuRIL, GPT-2) on a custom Urdu dataset, incorporating sentiment analysis, embedding generation, and domain-specific language processing tailored for mental health support.
- Engineered the complete Python backend for information retrieval and collaborated on a Flutter-based front-end interface, delivering an accessible and seamless experience for native Urdu-speaking users.

Virtuality Labs, Information Technology University — Lahore, Pakistan

Lab Lead - Natural Language Processing & Virtual Reality *2023 – 2024*
Supervisor: Ibrahim Ghaznavi (ITU), Abid Hussain (Copenhagen Business School)

- Mentored **17 interns**, managed international research collaborations with institutions including Imperial College London, Copenhagen Business School, Toronto Metropolitan University, and Kristiania College Norway, and represented the lab in demonstrations to government delegations including the Chief Minister of Punjab.

Project: First-ever VR Classroom Field Experiment in Pakistan ***ECIS 2025*** | Paper

- Volunteered and Led a team of junior researchers in designing and conducting a large-scale field experiment comparing VR, Face-to-Face, and Zoom teaching conditions across 100+ first-year CS students.

Intelligent Machines Lab (IML), Information Technology University — Lahore, Pakistan

Research Intern — 3D Computer Vision *Aug. 2023 – Apr. 2024*

Supervisor: Mohsen Ali (ITU & University of Florida)
Project: Real-time 3D Reconstruction & Neural Rendering — Joint IML–ETRI (South Korea) Research Project

- Developed a large-scale synthetic 4D dataset (∼2TB) comprising 80+ high-fidelity indoor scenes in Unity3D using the Perception Toolkit, incorporating physics-based animations and multimodal outputs (RGB-D, segmentation, video).
- Benchmarked temporal reconstruction pipelines using SfM, COLMAP, NeRF, and 3D Gaussian Splatting with iterative refinement, focusing on neural radiance fields and temporal view synthesis for optimal reconstruction quality.

Information Technology University — Lahore, Pakistan

Teaching Assistant *Aug. 2023 – June 2024*

- **CS152** — Object Oriented Programming (Spring 2022)
- **CS104** — Application of Information and Communication Technologies (Fall 2023)
- **CS595** — Theory and Application of Virtual Reality (Spring 2024)

Additional Projects

Fine-tuned LLaMA 2(7B) for "Friends" Screenplay Scripts Generation | *Python, PyTorch, Transformers, LoRA*

- Fine-Tuned *LLaMA 2(7B)* LLM on 236 episodes to generate new scripts by episode name.
- Used QLoRA for efficient 4-bit quantization and LoRA-based parameter-efficient tuning. Leveraged GPU acceleration, adaptive learning rates, and gradient checkpointing for supervised fine-tuning.

Gaussian Optimization for Image Reconstruction | *Python, PyTorch, NumPy, CV2* | Lab Lead: Usama Hassan | Code

- Co-developed a 2D Gaussian Splatting model for image reconstruction via GPU-accelerated backpropagation over mean and covariance tensors in PyTorch, inspired by the original 3D Gaussian Splatting paper, and predating subsequent works applying the same technique to compression and geometrically accurate radiance fields.

Photoshopped Images Detection | *Python, TensorFlow, Numpy, CV2, Matplotlib* | Advisor: Mehwish Ghafoor

- Fine-tuned Meta's ConvNeXt model on a dataset of 2,000+ images for detecting Photoshop-based face manipulations (Face Aware Liquify), achieving **65%** accuracy after 10 training epochs and outperforming average human detection rates of 53.5%.
- Conducted comparative analysis against Transformer-based models following the methodology of a ICCV 2019 paper.

Awards & Honors

National Grassroot ICT Research Initiative (NGIRI) 2023–24 | *IGNITE* 2023 – 2024

- Awarded the NGIRI 2023–24 national research grant (a federal ICT R&D fund), receiving **100,000 PKR** for *"Harnessing Mixed Reality and Drones for Agriculture using Generative AI"*, selected among outstanding final-year projects across Pakistani universities.

Undergrad Ehsaas Scholarship | *Higher Education Commission* 2020 – 2024

- Secured a highly competitive, fully-funded scholarship for the bachelor's based on history of academic excellence.

Google Competitive Programming Competition | *Google Developer Student Club, ITU* 2022

- Secured 4th position in a competitive programming among 150+ participants across the country.

Media & Press Coverage

- **National TV**: Research aired live on ARY News and Samaa TV.
- **Print & Online**: Featured in Daily Times and Academia Magazine.
- **Documentary**: Covered in an official ITU YouTube documentary titled *"VR Classes at ITU | 1st time in Pakistan"*.
- **Government**: Project demonstrated live to the Chief Minister of Punjab.

Extracurricular & Volunteering

Co-Founder & Head of Game Arts and Design Dec 2022 – Mar 2024
ITU Game Society | *Information Technology University* *Lahore, Pakistan*

- Co-founded the ITU Game Society and led all game arts and design operations, organizing game development workshops and eSports tournaments for 100s of students across the country.

Technical Skills

Languages: Python, C/C++, C#
Libraries: PyTorch, TF, NumPy, Pandas, Sklearn, Transformers, OpenAI Agents, LangChain/Graph, RAG, CV2, Matplotlib, Seaborn
Big Data & Dev: Hadoop, AWS, HTML & CSS, Ruby on Rails, Flutter, Flask, Git, Bash, Visual Studio
Simulation: Unity3D, Blender, OpenXR, OculusVR, DeepMind Control
Others: LLVMs, Adobe Suite, Office Suite, Systems Assembly

ACADEMIC SERVICES

Reviewer @ Neural Information Processing Systems (NeurIPS), 2025